

A REVIEW OF TWO-PHASE STEAM-WATER CRITICAL FLOW MODELS WITH EMPHASIS ON THERMAL NONEQUILIBRIUM

PRADIP SAHA

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THERMAL HYDRAULIC DEVELOPMENT DIVISION
DEPARTMENT OF NUCLEAR ENERGY, BROOKHAVEN NATIONAL LABORATORY
UPTON, NEW YORK 11973

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ABSTRACT

A review of the two-phase critical flow models has been presented with particular attention to the light water reactor (LWR) safety application. Pertinent experimental results have also been reviewed. No experiment has yet been performed in reactor-size pipe diameters (~ 300 mm) with relatively short pipe lengths (~ 1000 mm).^{*} From the small scale tests, it has been found that the critical flow rate increases rapidly as the pipe length is shortened to zero. This is particularly true if the upstream fluid condition is near saturation or subcooled, as is the case during the early stages of a hypothetical LOCA in a LWR system. In this case, both the homogeneous-equilibrium model (HEM) and the Moody model⁽¹⁰⁾ underpredict the critical flow rate data considerably. The effect of thermal nonequilibrium is believed to be the reason for this discrepancy.

Models which include the effect of thermal nonequilibrium have been reviewed in detail. Although a large number of lumped as well as distributed models have been proposed, there is no general correlation for any of these models. In view of the complexity of the problem, the relaxation-type models for the actual rate of vapor generation seem to be the most logical approach at this time.

^{*} Large-scale transient critical flow experiments have begun in Marviken test apparatus in Sweden. However, no detailed analyses of the tests are available yet.

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NOMENCLATURE

A	interfacial area density function defined by (59)
B	decay constant in (28) and (32)
C	contraction coefficient at pipe entrance
c	vapor mass concentration, $\alpha\rho_v/\rho_m$
c_p	specific heat at constant pressure
D	pipe inside diameter
F	function defined by (54)
G	mixture mass-flux
G_{cr}	critical mass-flux
g	acceleration due to gravity
HN	heterogeneous nucleation site density in liquid volume
h	specific enthalpy
h_{fg}	latent heat of vaporization
h_ℓ	heat transfer coefficient at vapor-liquid interface
K	Boltzmann constant
k	thermal conductivity
k	slip ratio, u_v/u_ℓ
L	pipe length
m	molecular weight
N	system characteristic parameter defined by (23)
N	bubble or droplet number density in mixture volume
N_ℓ	number of bubbles per unit volume of liquid
Nr_f	flashing relaxation number

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NOMENCLATURE
(Cont.)

Nu	Nusselt number
n	coefficient of polytropic expansion
P	pressure
Pe	Peclet number
r	bubble radius
r_0	initial bubble radius
r^*	critical pressure ratio
R_u	universal gas constant
S	specific entropy
T	temperature
T_{sat}	saturation temperature
t	time
u	velocity
v	specific volume
x	actual vapor flow quality
x_{eq}	equilibrium vapor flow quality
z	axial distance

Greek Symbols

α	vapor void fraction
β	function defined by (40)
Γ_v	actual mass rate of vapor generation per unit volume
Γ_{eq}	equilibrium vapor generation rate per unit volume

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NOMENCLATURE
(Cont.)

γ	ratio of specific heats
θ	contact angle
λ_e	arbitrary constant in (58)
ρ	density
σ	surface tension
τ	time constant
τ^*	nondimensional time constant in (65)
ϕ	contact angle function defined by (56)

Subscripts

e	exit
eq	equilibrium
f	liquid (saturated)
g	vapor (saturated)
HEM	homogeneous-equilibrium model
LT	long tube
l	liquid (general)
m	two-phase mixture
o	upstream or inlet
t	throat
v	vapor (general)

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1. INTRODUCTION

Fluid discharge rate through a pipe break is a key parameter in the analysis of hypothetical loss-of-coolant accident in nuclear reactor systems. It is expected that during most of the blowdown transient, the flow will be choked at the break. Consequently, the flow rate is expected not to depend on the downstream (containment) conditions, but on the upstream fluid- and thermodynamic conditions, piping size and system geometry. Therefore, to predict the discharge rate accurately for a LOCA situation, one must use a critical flow model which is applicable to large diameter pipes ($D \sim 300$ mm) of relatively short lengths, a wide range of upstream vapor qualities including subcooled liquid, and a wide range of upstream pressures.

The main purpose of this report is to review the presently available critical flow models, and try to identify a model which is best suited for the LOCA application of water reactors. First, the well-known homogeneous-equilibrium model is reviewed and its success/failure to predict the test data is discussed. Secondly, the models which attempted to incorporate the effect of relative velocity between the phases, but excluded the effect of thermal nonequilibrium, are discussed. This is followed by a brief review of experimental studies in short tubes and nozzles where the effect of thermal nonequilibrium is prominent. Finally, a detailed discussion of the models including the effects of thermal nonequilibrium is presented.

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2. HOMOGENEOUS-EQUILIBRIUM MODEL

2.1 Analytical Studies

Analytical and experimental work on two-phase critical flow has been in progress for the last three decades. Although early researchers^(1,2,3) noted the role of a metastable liquid state, the isentropic Homogeneous-Equilibrium Model (HEM) is still one of the more popular critical flow models. As the name implies, the model is based on the following assumptions:

- a. the flow is adiabatic and frictionless, hence, isentropic
- b. the liquid and vapor velocities are equal at any given point
- c. the flow is in thermodynamic equilibrium at every point.

Assuming a stagnant upstream condition, and utilizing assumption (a), one can express the mass velocity at any section as

$$G = \rho_m u_m = \rho_m [2g (h_0 - h)]^{1/2} . \quad (1)$$

For homogeneous flow,

$$\frac{1}{\rho_m} = \frac{1-x}{\rho_\ell} + \frac{x}{\rho_v} \quad (2)$$

and

$$h = (1-x) h_\ell + x h_v . \quad (3)$$

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On the other hand, for flow in thermal equilibrium,

$$T_{\ell} = T_v = T_{\text{sat}} (P)$$

so that

$$\begin{aligned} \rho_{\ell} &= \rho_f (P), & \rho_v &= \rho_g (P) \\ h_{\ell} &= h_f (P), & h_v &= h_g (P) \end{aligned} \quad (4)$$

Therefore, Equation (1) can be rewritten as

$$G = \frac{\{2g [h_o - (1 - x) h_f (P) - x h_g (P)]\}^{\frac{1}{2}}}{\frac{1 - x}{\rho_f (P)} + \frac{x}{\rho_g (P)}} \quad (5)$$

Invoking the assumption of isentropic flow, vapor quality at any section can be written as

$$x = \frac{S_o (P_o, h_o) - S_f (P)}{S_g (P) - S_f (P)} \quad (6)$$

The critical mass flux is then calculated from Equations (5) and (6) by finding the downstream pressure, P , for which the mass-flux, G , is maximum. It should be noticed that the critical mass-flux thus obtained, depends only on the upstream thermodynamic conditions (P_o and h_o), and not on pipe diameter or pipe length. Moody⁽⁴⁾ has recently mapped these critical mass-flux values for

steam-water system over a wide range of stagnation pressures and enthalpies including subcooled water. The map is reproduced here (Fig. 1) for completeness. This method provides a simple and direct method specially suited for hand calculations.

Some previous researchers⁽⁵⁻⁸⁾ preferred to express the critical mass-flux in terms of the fluid properties at the location of choking which usually occurs at the exit of a constant diameter pipe or at the throat of a converging-diverging nozzle. With the choking criterion

$$\left(\frac{\partial G}{\partial P_e} \right)_s = 0 \quad (7)$$

one can show that for isentropic flow

$$G_{\max}^2 = - \left(\frac{\partial P_e}{\partial v_e} \right)_s \quad (8)$$

For homogeneous, equilibrium flow

$$v_e = (1-x_{e,eq}) v_f + x_{e,eq} v_g \quad (9)$$

Therefore, Equation (8) becomes

$$G_{\max}^2 = - \frac{1}{x_{e,eq} \left(\frac{\partial v_g}{\partial P_e} \right)_s + (v_g - v_f) \left(\frac{\partial x_{e,eq}}{\partial P_e} \right)_s + (1-x_{e,eq}) \left(\frac{\partial v_f}{\partial P_e} \right)_s} \quad (10)$$

A graphical representation of the above equation for steam-water system is

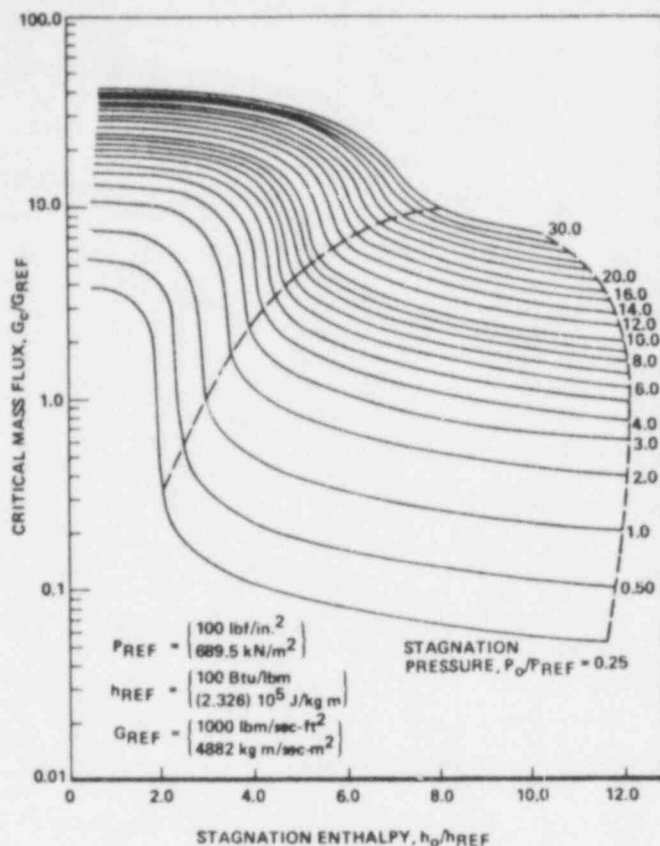


Figure 1. Steam-water critical mass-flux according to the homogeneous-equilibrium model (in terms of upstream conditions). (Moody⁴) (BNL 5-1230-78)

taken from Reference 8 and reproduced in Figure 2. It should be noted that the pressure and the equilibrium vapor quality (or total energy) at the critical (choking) point must be known to be able to determine the maximum (critical) mass-flux at that point. This is a major disadvantage for "a priori" hand calculation, but does not seem to be an obstacle to marching technique solutions generally typical of computational methods.

2.2 Experimental Studies in Long Tubes and Comparison With HEM

Systematic experimental studies on two-phase, steam-water critical flows

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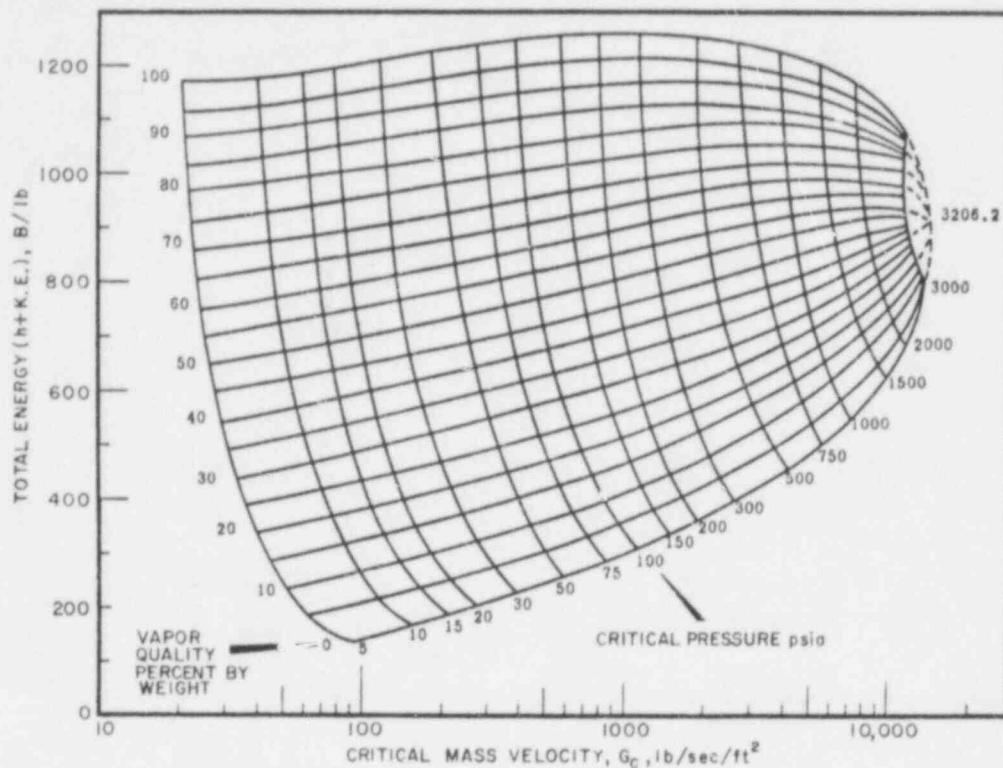


Figure 2. Steam-water critical mass-flux according to the HEM (in terms of critical conditions). (Zaloudek⁸) (BNL 5-1431-78)

over a wide range of equilibrium qualities were apparently started by Isbin, Moy and Da Cruz⁽⁵⁾ in University of Minnesota. Critical flow rates were measured in constant diameter pipes of 0.3743" to 1.0425" (9.5 to 26.5 mm) in inside diameter and 2 ft. (610 mm) in length. Pressures at the exit of the test pipes ranged from 4 to 43 psia (0.28 to 2.96 bar), whereas the exit equilibrium quality ranged from 0.01 to 1.0. In general, the experimental values of the critical flow rates were much higher than those calculated by the homogeneous-equilibrium model based on the exit fluid properties. The discrepancy between the experimental

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values and the predicted values was maximum as the exit equilibrium quality approached zero and it was minimum (almost zero) as the exit equilibrium quality approached unity. No explanation for this discrepancy was offered. Also, no specific effect of tube diameter could be seen from the data.

Fauske^(6,7) extended the work of Isbin, et al.,⁽⁵⁾ at higher critical pressures (2.76 to 24.8 bar). Smaller tube diameters (3.18, 6.83, and 12.25 mm), but longer tube lengths (1429 and 2794 mm) were employed. The exit equilibrium qualities ranged from 0.01 to 0.7. The data on the critical flow rate showed the same behavior with respect to exit qualities as the earlier data taken by Isbin, et al.,⁽⁵⁾ and the homogeneous-equilibrium model underpredicted the data considerably.

Zaloudek⁽⁸⁾ performed similar experiments in 0.520" and 0.625" (13.2 and 15.9 mm) diameter tubes of 24" to 48" (610 to 1219 mm) in length. Exit pressures were measured more accurately than the previous workers by locating a pressure tap only 1/32" (0.8 mm) from the exit. Critical pressures ranged from 40 to 110 psia (2.76 to 7.58 bar), whereas the exit equilibrium quality ranged from 0.004 to 0.99. The critical flow data fell slightly above the data of Isbin, et al.,⁽⁵⁾ but showed the same general trend as the other data. No effect of tube diameter and/or tube length could be detected.

Faletti and Moulton⁽⁹⁾ measured the exit pressure most accurately by inserting a concentric pressure probe of 0.187" (4.75 mm) or 0.375" (9.53 mm) outside diameter into a tube of 0.574" (13.58 mm) inside diameter. Thus the test section had a hydraulic diameter of 0.387" (9.83 mm) or 0.199" (5.05 mm). The length of the test section was varied from 9" to 35.14" (229 to 893 mm). A short test section of 0.531" (13.5 mm) in length and 0.199" (5.05 mm) in hydraulic diameter was also used. Critical pressures ranged from 25 to 80 psia

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(1.72 to 5.52 bar), and the exit equilibrium qualities ranged from 0.001 to 0.96. The data showed very good agreement with Zaloudek's data⁽⁸⁾ and they fell above the data of Isbin, et al.⁽⁵⁾ This showed the sensitivity of the critical flow rate on the critical pressure, and the importance of measuring the latter accurately. The general behavior of the data was the same as other previous data and no length or diameter effect for test sections above 9" (229 mm) in length was noticed. On the other hand, the short test section of 0.5313" (13.5 mm) in length exhibited critical flow rates ten times those predicted by the homogeneous-equilibrium model. However, it was not clear whether those high flow rates were caused by the short tube length or by the low vapor quality (~ 0.001) or by both.

James⁽⁴⁵⁾ conducted steam-water critical flow experiments in pipes of diameter 3", 6", and 8" (76, 152, and 203 mm). The length of the discharge pipes was 12 ft. (3658 mm) and the exit pressure tap was located at 0.25" (6.4 mm) from the end of the pipe. The stagnation enthalpies ranged from 230 to 1200 Btu/lbm (5.35×10^5 to 27.91×10^5 J/Kg) and the critical, i.e., exit, pressures ranged from 14 to 54 psia (0.97 to 4.4 bar). A geothermal bore in New Zealand was used as the source of steam-water mixture. The experimental critical mass-fluxes were approximately 10% larger than those obtained by Isbin, Moy and Da Cruz.⁽⁵⁾ Therefore, the experimental values were also larger than the predictions of the homogeneous-equilibrium model. James was able to correlate the mass-flux data in terms of critical pressure and stagnation enthalpy only, and no effect of pipe diameter was found.

From the above discussion of the experimental studies it is evident that one cannot rely on the homogeneous-equilibrium model for an accurate prediction of the critical flow rate. Therefore, a fresh look to the assumptions of the HEM

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was needed. It was found that except for high length-to-diameter ratios, the pressure drop across a tube was caused mainly by the acceleration of the fluid and not so much by the friction. Therefore, the assumption of isentropic expansion of fluid could be retained for adiabatic systems. It was also argued that unless the tube was short, the assumption of thermodynamic equilibrium between the two phases would hold good. Therefore, attention was focused to relax the assumption of homogeneous flow and try to accommodate the effect of relative velocity between the phases in a theoretical model.

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3. NONHOMOGENEOUS, EQUILIBRIUM MODELS

Fauske^(6,7) led the field in developing a new model. He assumed that in addition to the usual choking criterion of $(\partial G / \partial P_e)$ equal to zero, the pressure gradient attained a finite, maximum value at the location of choking. Considering the momentum equation, Fauske found it necessary to define the specific volume of two-phase mixture as

$$v \equiv \frac{x^2 v_g}{\alpha} + \frac{(1-x)^2 v_f}{(1-\alpha)} \quad (11)$$

where α was the vapor void fraction. It should be noted that the above definition of specific volume for a mixture is not correct, and is known as the "momentum" specific volume in literature. By introducing a slip ratio, k , i.e., the ratio of average vapor velocity to average liquid velocity, the "momentum" specific volume (11) can be written as

$$v = \frac{1}{k} \left[(1-x) v_f k + x v_g \right] \left[1 + x (k-1) \right] \quad (12)$$

Finally, the assumption of maximum pressure gradient at the critical condition led to

$$\frac{\partial v}{\partial k} = x(1-x) \left(v_f - \frac{v_g}{k^2} \right) = 0 \quad (13)$$

so that for critical flow, the slip ratio at the critical location became

$$k = \sqrt{\frac{v_g}{v_f}} \quad (14)$$

Equations (12) and (14) were then used to calculate the critical flow rate from

$$G_{\max}^2 = - \left(\frac{\partial P_e}{\partial v_e} \right) . \quad (15)$$

The above derivative was evaluated by following isentropic as well as isenthalpic processes, and the difference was found to be minor. Fauske's model predicted larger critical flow rates than the homogeneous-equilibrium model predictions previously discussed, and showed good agreement with the data of Isbin, et al., Fauske, Zaloudek, and Faletti and Moulton, described earlier. It also showed good agreement⁽⁴⁶⁾ with the large diameter pipe data of James.⁽⁴⁵⁾

Starting from continuity and energy equations, and by assuming an isentropic process, Moody⁽¹⁰⁾ was able to express the local mass-flux in terms of stagnation properties (entropy and enthalpy), local slip ratio, k , and local pressure, P . Assuming the slip ratio, k , and pressure, P , as independent variables, Moody defined the critical condition as

$$\left(\frac{\partial G}{\partial k} \right)_P = 0 \quad (16)$$

and

$$\left(\frac{\partial G}{\partial P} \right)_k = 0 . \quad (17)$$

Equation (16) led to the following slip ratio for critical condition:

$$k = 3 \sqrt{\frac{v_g}{v_f}} . \quad (18)$$

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Applying the above equation in the expression for local mass-flux and by searching the critical pressure P which would satisfy condition (17), Moody was able to map the maximum steam-water flow rate in terms of stagnation properties. Maximum flow rates in terms of local (critical) static pressures and equilibrium vapor qualities were also mapped. Like Fauske's model, this model also predicted flow rates which were higher than the predictions of the homogeneous-equilibrium model. However, when compared to the data of Isbin, et al.,⁽⁵⁾ Fauske,^(6,7) Zaloudek,⁽⁸⁾ and Faletti,⁽⁹⁾ the model slightly overpredicted the measured flow rates for exit equilibrium qualities between 0.01 and 0.5, and underpredicted the data slightly for qualities between 0.5 and 1.0.

The discrepancy between the models of Fauske^(6,7) and Moody⁽¹⁰⁾ becomes obvious from Equations (14) and (18). Both claimed to maximize the mass-flux by using two different values of slip ratios at the critical section. A closer look towards the models revealed that the Fauske model minimized the so-called momentum specific volume, whereas the Moody model minimized the so-called kinetic energy specific volume. That is to say that the Fauske model maximized the momentum flow rate, whereas the Moody model maximized the kinetic energy flow rate. Moreover, Cruver and Moulton⁽¹¹⁾ pointed out that Fauske's critical slip ratio did not necessarily maximize the pressure gradient dP/dZ at the critical location. They concluded that the pressure gradient in the two-phase critical flow would not attain a finite maximum because of a variation in slip ratio.

Levy⁽¹²⁾ used continuity and two momentum equations, one for each phase, to derive a relationship between the vapor quality and the vapor void fraction. Assuming an isentropic process, he then expressed the critical mass-flux as

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$$G_{\max}^2 = \left(\frac{\partial P}{\partial v_M} \right)_s \quad (19)$$

where v_M is identical to Fauske's so-called momentum specific volume defined by Equation (11). It is interesting to note that at low pressure, Levy's slip ratio can be approximated by⁽¹³⁾

$$k = \sqrt{\frac{v_g}{v_f}} \sqrt{2 \alpha} \quad (20)$$

Because of the similarity between Equation (14) and (20), Levy's prediction for critical mass-flux matched closely with that of Fauske, and agreed reasonably well with the data of Isbin, et al.,⁽⁵⁾ Fauske,^(6,7) Zaloudek,⁽⁸⁾ and Faletti.⁽⁹⁾ Although the models of Fauske, Moody, Levy were successful in predicting critical flow rate data at low pressures (< 25 bar) and high length-to-diameter ratios (> 20), it must be kept in mind that none of the above models was completely correct in its theoretical treatment of the problem. Some of the defects of Fauske's and Moody's models have been pointed out earlier. It should be noted that analysis of Levy was in error due to a wrong definition of mixture entropy. Moody also used the same definition for mixture entropy which is valid only for the homogeneous, equilibrium flow. However, the variation in slip ratios changes the resultant critical flows slightly, as evidenced by the Fauske,^(6,7) Moody,⁽¹⁰⁾ and Levy⁽¹²⁾ model predictions.

The preceding sections have all dealt with long tubes and pipes relative to the diameter ($L/D > 20$). The net conclusion is that nonhomogeneous slip models based on equilibrium assumptions do a reasonably good job for pipes with

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large length-to-diameter ratios ($L/D > 20$). In the following section the experiments conducted in test sections of small length-to-diameter ratios will be described. This will reveal the inadequacy of the equilibrium models and show the importance of incorporating the effect of thermal nonequilibrium in critical flow models.

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4. EXPERIMENTAL STUDIES IN SHORT TUBES AND NOZZLES

It was well known for a long time that flow rates of two-phase mixture or saturated liquid through orifices and short nozzles were much higher than the HEM predictions.⁽¹⁻³⁾ However, Zaloudek⁽¹⁴⁾ apparently started a systematic investigation of critical flow of initially subcooled water through short tubes of length 0.03" to 5" (0.8 to 127 mm) and with sharp entrance edges. The tube diameters were 0.25", 0.50" and 0.625" (6.35, 12.7 and 15.9 mm), and the upstream pressure ranged from 100 to 350 psig (7.9 to 25.1 bar). Two types of choking phenomena were observed by Zaloudek. The first type occurred near the vena-contracta at the inlet of the test piece when localized flashing occurred as the pressure dropped below the corresponding saturation pressure. The receiver (back) pressure was still higher than the saturation pressure. As the back pressure was decreased, a free discharge-type flow pattern appeared, and the flow rate increased. When the back pressure fell well below saturation, a second type of choking occurred at the exit of the test piece. It is the second type of choking which is more important from the practical point of view and will be discussed here.

Critical mass-fluxes corresponding to the second type of choking were considerably higher than the predictions of the homogeneous-equilibrium model, but were in agreement with Burnell's surface tension model⁽²⁾ and Bailey's surface evaporation model.⁽³⁾ Visual observation in a 12.7 mm diameter and 31 mm long glass tube revealed that at the critical condition a vapor annulus surrounded a liquid core with some vaporization near the exit. The liquid core was believed to be metastable although no temperature measurement was taken. Variation of tube diameter and the tube length showed only a very slight effect on the critical mass flux, and no conclusion could be drawn because of the scatter of the

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data. The dissolved gas in water was partially blamed for the scatter, but no systematic observations of this parameter were reported. It was observed that a reduction in the amount of dissolved gas, by heating the water to 90°C in an open tank, could increase the critical mass flux by 5 to 8%.

Zaloudek⁽¹⁵⁾ extended the above investigation to higher pressures in a tube of diameter 0.505" (12.8 mm) and a length of 10" (254 mm). Thus the length-to-diameter ratio of the test section was approximately twenty. There was also a 20° conical approach section. Upstream pressures ranged from 400 to 1800 psia (27.6 to 124 bar), whereas upstream enthalpies varied from 420 to 533 Btu/lbm (9.76×10^5 to 12.39×10^5 J/kg). It was found that Fauske's model^(6,7) predicted the critical flow rate data quite well when two-phase mixture entered the test section. However, when subcooled water entered the test section, the critical flow rates were much higher than those predicted by the Fauske model.^(6,7) This discrepancy was indeed attributed to the thermal nonequilibrium phenomenon which retarded the vapor generation in the tube and produced lower vapor qualities at the exit.

Fauske⁽¹⁶⁾ studied the effect of test section length-to-diameter ratios for saturated water discharging from high pressure sources (6.9 to 124 bar). The test section inside diameter was 0.25" (6.35 mm) & the length was varied from 0 to 254 mm (i.e., $0 < \frac{L}{D} < 40$). All the tubes had sharp-edged entrances. For $\frac{L}{D}$ ratios higher than 12, the critical flow rates agreed well with the Fauske model.^(6,7) On the other hand, for $\frac{L}{D}$ values less than 3, the data were highly underpredicted by the Fauske model, but could be well predicted by the following incompressible flow equation for the orifice:

$$G = 0.61 \sqrt{2 \rho_L g (P_0 - P_c)} \quad (21)$$

where P_c was the pressure at the exit which yielded maximum flows. Therefore, Fauske concluded that for small $\frac{L}{D}$ (< 3), the fluid broke immediately from the wall and remained as a metastable liquid core jet with some evaporation from the jet surface. For the intermediate range of $\frac{L}{D}$ ratios (3 to 12), however, the measured flow rates fell between Equation (21) and the Fauske model. Therefore, it appeared that a breakup of metastable liquid jet took place in that intermediate range. The experimental results of this study are shown in Figure 3.

Uchida and Nariai⁽¹⁷⁾ performed similar tests at low pressures (1.96 to 7.85 bar) in copper, brass and glass tubes of 4 mm inside diameter. The tube length was varied from 0 to 2500 mm, (i.e., $0 < \frac{L}{D} < 625$). The tube had a sharp-edged entrance through which initially saturated or subcooled water flowed to the atmosphere. The flow rates were found to decrease with increasing tube length and showed only fair agreement with Levy, Fauske, or Moody's model for long tubes. Flow rates for subcooled water were always higher than those for saturated water at the same upstream pressure. However, the data from this test were in qualitative agreement with Fauske's test,⁽¹⁶⁾ and visual observation (in a glass tube) showed that even for saturated upstream condition and long tube (1410 mm), phase change started at the middle of the tube. Therefore, the liquid did become superheated in the glass tube which might have had less nucleation sites than a metallic pipe.

Starkman, et al.⁽¹⁸⁾ used two convergent-divergent nozzles of throat diameters 0.438" (11.1 mm) and 0.252" (6.4 mm) to measure critical flow rates of steam-water mixtures at upstream pressures of 100 to 1000 psia (6.9 to 69 bar) and upstream vapor qualities of zero (no subcooling) to 20%. The flow rates were found to increase rapidly as the upstream quality approached zero. Over the entire range of the experiment, the flow rates were higher than the predictions

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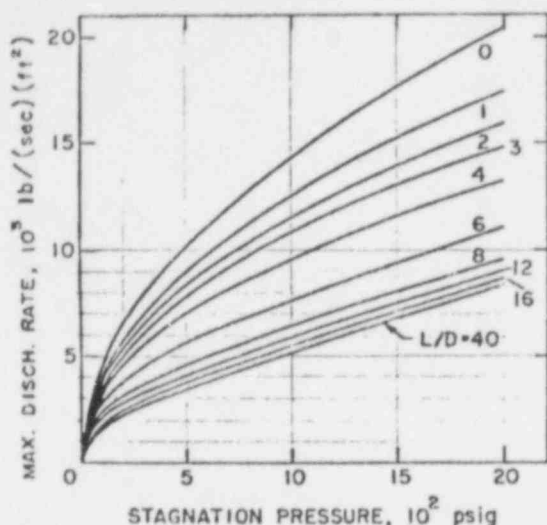


Figure 3. Maximum discharge rate of saturated water through a 6.35 mm I.D. tube. (Fauske¹⁶) (BNL 5-1227-78)

of the homogeneous-equilibrium model, and the deviation between the data and the prediction reduced to 10% as the upstream quality was raised above 10%. No systematic effect of throat diameter was noticed.

Starkman, et al., compared their data also with a "Homogeneous Frozen" model (HFM). This is a limiting model based on the following assumptions:

- a. The velocities of both phases are equal, i.e., homogeneous flow.
- b. No heat or mass transfer takes place between the phases; thus, the vapor quality remains constant throughout the channel.
- c. The vapor expands isentropically; i.e., $Pv_g^\gamma = \text{constant}$.
- d. The kinetic energy of the stream evolves entirely from the vapor expansion due to pressure drop and area change.
- e. The critical flow rate occurs when the vapor Mach number reaches unity at the throat.

The final expression for the critical mass-flux at the throat becomes^(18,20)

$$G_{\text{crit}} = \frac{1}{x_0 v_{g,0} (r^*)^{-1/\gamma} + (1-x_0) v_{f,0}} \times \left[2g x_0 v_{g,0} P_0 \frac{\gamma}{\gamma-1} \left\{ 1 - (r^*)^{\frac{\gamma-1}{\gamma}} \right\} \right]^{\frac{1}{2}} \quad (22)$$

where the pressure ratio, r^* , is given by:

$$r^* = \frac{P_t}{P_0} = \left[\frac{2}{\gamma+1} \right]^{\frac{\gamma}{\gamma-1}}$$

The adiabatic coefficient of expansion, γ , for steam is usually taken as 1.3.

It was found that for upstream qualities greater than approximately 2%, the critical flow rate data of Starkman, et al.,⁽¹³⁾ could be bounded by the homogeneous-equilibrium and the homogeneous-frozen models. This is shown in Fig. 4. At very low qualities the frozen model predicts much lower flow rates which do not agree with the trend of the data. The reason for this anomaly is that the kinetic energy of the liquid phase was ignored completely.

Schrock, et al.,⁽¹⁹⁾ extended the above study to initially subcooled water. They also employed a third nozzle of throat diameter 0.156" (3.96 mm). The stagnation pressure ranged from 100 to 1300 psia (6.9 to 89.6 bar), and the subcooling ranged from 0 to 60°C. The critical flow rate data always fell between the homogeneous-equilibrium model and the all liquid, i.e., Bernoulli, flow model. Existence of metastable state of water was recognized, and a two-step frozen model was advanced. According to this model, the flow remains all liquid until the pressure drops below the corresponding saturation pressure by a certain amount. At this point, the flow suddenly attains the equilibrium condition. However, beyond that point the two-phase mixture is again assumed to flow according to the homogeneous-frozen model. This two-step model showed only limited success, and no correlation for the point of discontinuity was proposed.

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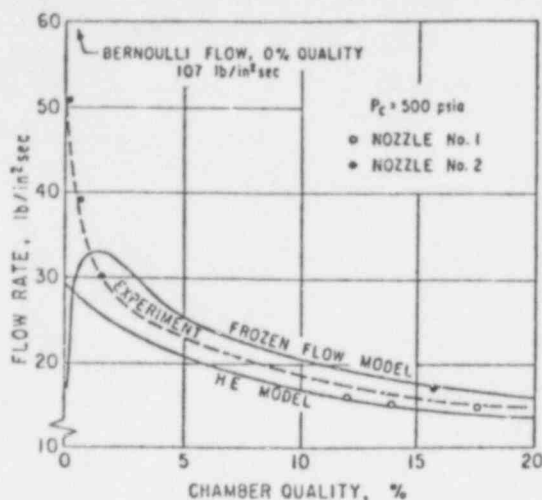


Figure 4. Comparison of experimental flow rates with the homogeneous-equilibrium and homogeneous-frozen models. (Starkman, et al.¹⁸) (BNL 5-1224-78)

Deich, et al.,⁽²⁰⁾ measured critical flow rate at low pressure (1.22 bar), but higher upstream qualities (20 to 100%) in a convergent-divergent nozzle of throat diameter 32.55 mm. Their experimental data on flow rates were always higher than those predicted by both the homogeneous-equilibrium and the homogeneous-frozen models. The reason for this apparent contradiction with the data of Starkman, et al.,⁽¹⁸⁾ was not given.

Recently, Sozzi and Sutherland⁽²¹⁾ reported critical flow rate data in a variety of nozzles. Most of the data were taken in (1) a convergent-divergent nozzle with well-rounded entrance and 0.5" (12.7 mm) diameter throat, (2) a convergent nozzle with well-rounded entrance and 0.5" (12.7 mm) diameter exit section, i.e., the same configuration as Nozzle No.1 without the divergent section, (3) a sharp-edged orifice of diameter 0.5" (12.7 mm), and (4) a convergent nozzle, the same as Nozzle No. 2, but with 0.75" (19 mm) exit diameter. Limited amount of data were also taken in convergent-divergent nozzles of throat diameters 2.125" (54 mm), 3.0" (76.2 mm) and 1.1" (28 mm). In addition, constant diameter (12.7 mm) pipes of various length were attached to Nozzle No. 2 and 3 to study the effect of pipe length on critical flow. Stagnation pressure was varied from 600 to 1000 psia (41.4 to 69 bar) and the stagnation temperature ranged from 450 to 550°F (232 to 288°C).

Several interesting observations were made. Like other investigations, the critical flow rate in this study was also found to decrease with increasing upstream quality.* As expected, a well-rounded convergent nozzle (No. 2) allowed more flow than a sharp-edged orifice of the same diameter. However, when a divergent section was added to a convergent nozzle, the critical flow rate decreased considerably. The flow rate was found to decrease further when the divergent section was replaced with a constant diameter pipe of the same length. The pipe diameter, of course, was equal to the throat diameter. Therefore, it could be concluded that the geometry (or, the choking location) played an important role in critical flow phenomenon, particularly for short flow lengths.

Sozzi and Sutherland⁽²¹⁾ also found that the critical mass flux decreased with increasing throat diameter. This was clearly shown for convergent nozzles of two different throat diameters (12.7 and 19 mm). The same trend was shown for convergent-divergent nozzles with various throat diameters (12.7, 28, 54 and 76.2 mm). These findings were in contradiction with earlier investigations of Starkman, et al.,⁽¹⁸⁾ but in agreement with the studies of Bryers and Hsieh.⁽⁴⁷⁾

The effect of pipe length was similar to that found by Fauske⁽¹⁶⁾ and Uchida and Nariai.⁽¹⁹⁾ The critical flow rate decreased sharply as the pipe length was increased. However, above a pipe length of 5" (127 mm), only a small decrease in critical flow rate was observed, and the flow rates in these longer pipes seemed to agree well with the homogeneous-equilibrium model prediction. Sozzi and Sutherland, therefore, concluded that the effect of thermal non-equilibrium could be neglected if the flow lengths were larger than 5" (127 mm). This seems to be quite arbitrary, and may not be valid for larger pipe diameters.

*Quality in this study was based on mixture density, not on mixture enthalpy. Therefore, this quality, when positive, is the same as the static quality, or the vapor mass concentration.

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Morrison⁽²²⁾ provided further data in support of the above conclusion of Sozzi and Sutherland. He took critical flow data in a well-rounded, convergent nozzle of 1.1" (28 mm) exit diameter with a 5.25" (133 mm) long constant diameter (28 mm) section attached to it. The data seemed to agree well with the HEM prediction except for zero upstream quality. Morrison also showed that the critical flow rate for the above test section was smaller than that for a convergent-divergent nozzle of the same throat diameter and same overall length. He attributed this difference to the nonequilibrium effects associated with the convergent-divergent nozzle, or in other words, to the different locations of choking. Moreover, no diameter effect was found when constant diameter (12.7 or 28 mm) pipes of at least 127 mm lengths were attached to convergent nozzles of 12.7 or 28 mm exit diameter. This was partially in agreement with other investigations.^(5,8,9)

Hutcherson⁽²³⁾ ran transient blowdown tests from a vessel with internal skirt. The discharge pipe was 4.257" (108 mm) in inside diameter and 12.755" (324 mm) in length. Thus, the length-to-diameter ratio of the discharge pipe was approximately three. The pipe was connected to the vessel through a short conical convergent section. Initially saturated water at 290 and 402 psia (20 and 27.7 bar) was discharged through the pipe, and conditions (pressure and quality) at the entrance of the pipe were calculated as a function of time. Based on these upstream conditions, critical discharge flow rates were calculated assuming the HEM and a nonequilibrium model developed by Henry and Fauske,⁽²⁸⁾ discussed later. The nonequilibrium model, which predicted higher flow rate than the HEM, showed better agreement with the measured flow rate during the early part of the transient when the flow was believed to be choked. Therefore, it is not certain that the nonequilibrium effect is negligible for large diameter pipes (> 100 mm) if the pipe length exceeds 127 mm as suggested by Sozzi and Sutherland.⁽²¹⁾

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In summary, it can be said that the critical mass-flux is a strong function of pipe length, particularly for short pipes and low upstream qualities including the subcooled liquid. Under these circumstances, the measured critical flow rates, which also depend on the entrance geometry, have been found to be much higher than the predictions of the homogeneous-equilibrium model. In some cases,^(15,16,17) the flow rates were even higher than the equilibrium, slip-flow models of Fauske,^(6,7) Levy⁽¹²⁾ or Moody.⁽¹⁰⁾ Thermal nonequilibrium between the phases has been cited as the possible reason for this discrepancy. However, it has not yet been established whether the pipe length, L , or the length-to-diameter ratio, $\frac{L}{D}$, or both should be the governing parameter(s) in relation to the nonequilibrium effects. In Table 1, the experimental studies reviewed in this report are presented and the conditions where thermal nonequilibrium is important have been indicated. In the following section, the models which have attempted to take into account the effect of thermal nonequilibrium are discussed in detail.

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Table 1: Summary of Steam-Water Critical Flow Data in Constant Area Pipes

Researchers (Ref.)	System Conditions	Entrance Geometry	Pipe Geometry			Model Comparison		Remarks
			D (mm)	L (mm)	L/D	HFH	Equilibrium slip (6,7,10,12)	
Isbin, et al. (5)	$P_e = 0.28$ to 3 bar $x_{e,eq} = 0.01$ to 1.0	conical	9.5 to 26.5	610	23 to 64	poor except $x_{e,eq} \approx 1$	good	Exit pressure measurements may be in error.
Fauske (6,7)	$P_e = 3$ to 25 bar $x_{e,eq} = 0.01$ to 0.7	conical and sharp-edged	3.2 to 12.3	1429 and 2794	116 to 873	-do-	-do-	-do-
Zaloudek (8)	$P_e = 2.76$ to 7.58 $x_{e,eq} = 0.004$ to 0.99	conical	13.2 and 15.9	610 to 1219	38 to 92	-do-	-do-	Better exit pressure mea- surement.
Faletti and Moulton (9)	$P_e = 1.72$ to 5.52 bar $x_{e,eq} = 0.001$ to 0.96	conical	5 and 9.8	229 to 893	23 to 178	poor except $x_{e,eq} \approx 1$	good	Annular test section. Hy- draulic diame- ters indicated.
	-do-	conical	5	13.5	2.7	poor	poor	Nonequilibrium effects are im- portant.
James (45)	$P_e = 1$ to 4.4 bar $h_o = 5.35 \times 10^5$ to 27.9×10^5 J/kg	not reported	76 to 203	3658	18 to 48	poor	good	Upstream pres- sures not re- ported.

Table 1: Summary of Steam-Water Critical Flow Data in Constant Area Pipes (Cont.)

Researchers (Ref.)	System Conditions	Entrance Geometry	Pipe Geometry			Model Comparison		Remarks
			D (mm)	L (mm)	L/D	HEM	Equilibrium slip (6,7,10,12)	
Zaloudek (14)	$P_0 = 8$ to 25 bar subcooled liquid	sharp-edged	6.35 to 15.9	0.8 to 127	0.05 to 20	poor	poor	Nonequilibrium effects are im- portant. Ef- fects of dis- solve gases are also important.
Zaloudek (15)	$P_0 = 27.6$ to 124 bar $h_0 = 9.76 \times 10^5$ to 12.39×10^5 J/kg	20° conical	12.8	254	20	poor	poor, except 2-phase entrance	Nonequilibrium effects im- portant for subcooled liq- uid entrance.
Fauske (16)	$P_0 = 7$ to 124 bar $h_0 = h_{f,0}$	sharp-edged	6.35	0 to 254	0 to 40	poor, except $L/D > 12$	poor, except $L/D > 12$	Nonequilibrium effects im- portant for $L/D < 12$.
Uchida and Nariai (17)	$P_0 = 2$ to 8 bar $h_0 \leq h_{f,0}$	sharp-edged	4	0 to 2500	0 to 625	poor, except $L > 1000$ mm	poor, except $L > 1000$ mm	Significant nonequilibrium effects.
Sozzi and Sutherland (21)	$P_0 = 41$ to 69 bar $T_0 = 232$ to 288 C	well-rounded and sharp-edged	12.7	0 to 1775	0 to 140	poor, except $L > 127$ mm	comparison not shown	Significant nonequilibrium effects for $L < 127$ mm.
Morrison (22)	$P_0 = 69$ bar $x_0 = \frac{v - v_f(P_0)}{v_{fg}(P_0)}$ $= -0.002$ to 0.005	well-rounded	28	133	4.75	good, except $x_0 \approx 0$	-do-	Nonequilibrium effects are claimed to be insignificant because $L >$ 127 mm.

Table 1: Summary of Steam-Water Critical Flow Data in Constant Area Pipes (Cont.)

Researchers (Ref.)	System Conditions	Entrance Geometry	Pipe Geometry			Model Comparison		Remarks
			D (mm)	L (mm)	L/D	HEM	Equilibrium slip (6,7,10,12)	
Hutchenson (23) (Transient Blowdown)	Initial con- ditions: $P_o = 20$ and 27.7 bar $h_o = h_{f,o}$	conical	108	324	3	poor	poor	Significant nonequilibrium.

5. NONEQUILIBRIUM MODELS

5.1 Lumped Models

Henry, et al.,⁽²⁴⁾ recognized the fact that the effect of thermal non-equilibrium was to maintain the value of the actual vapor quality, x , below the corresponding equilibrium quality, x_{eq} . They defined a system characteristic parameter, N , such that

$$N \equiv \frac{x}{k x_{eq}} \quad (23)$$

where k is the slip ratio. Because $x < x_{eq}$, Nk must be less than unity. Henry, et al., also recognized, from their experimental studies in long tubes $\left(\frac{L}{D} > 40\right)$ ⁽²⁵⁾ and studies of Klingebiel and Moulton,⁽²⁶⁾ that the actual slip ratios at the choking plane were much lower than the values taken in Fauske's $(\sqrt{v_g/v_f})$ or Moody's $(\sqrt[3]{v_g/v_f})$ models. Therefore, they were able to derive the following expression for the critical mass-flux for low exit equilibrium qualities ($0 < x_{eq} < 0.02$):

$$G_{cr}^2 = \left\{ \frac{1}{\frac{N}{G_{cr,HEM}^2} - v_g x_{eq} \frac{dN}{dP}} \right\}_e \quad (24)$$

where $G_{cr,HEM}$ represents the critical mass-flux value under the homogeneous-equilibrium model and the right-hand side of Equation (24) is to be evaluated at the exit (or throat) conditions.

For low vapor qualities one can write

$$x = k \frac{\alpha}{(1-\alpha)} \frac{v_l}{v_g} = Nkx_{eq} \quad (25)$$

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Henry, et al.,⁽²⁴⁾ calculated the values of N from their experimental studies,⁽²⁵⁾ and assuming N to be a function of equilibrium quality only, they developed the following correlation:

$$\begin{aligned} N &= 20 x_{eq} & (0 < x_{eq} < 0.05) \\ N &= 1 & (x_{eq} > 0.05) \end{aligned} \quad (26)$$

The above correlation implicitly assumes that equilibrium is reached at 5% equilibrium-quality for a slip-ratio value of unity. This seems to have no physical basis. It was also found that at the choking plane $\frac{dN}{dP}$ could be taken to be zero. Therefore, Equation (24) reduces to:

$$G_{cr} = \frac{G_{cr,HEM}}{\sqrt{N_e}} \quad (27)$$

Henry, et al., applied Equation (27) to their data at moderate exit pressures (2.76 to 10.3 bar) and low exit equilibrium qualities (< 0.02) taken in long tubes ($\frac{L}{D} > 40$), and showed good agreement. Scatter at very low qualities was attributed to the dissolved gas in water used for the experiment. Notice that one has to know the exit pressure, which is usually not known, to be able to calculate the critical flow from the above model.

Henry⁽²⁷⁾ also proposed a model for initially subcooled or saturated water discharging from a long tube having a sharp-edged entrance. He assumed that the vapor generation in pipes of $\frac{L}{D}$ ratio less than 12 could be neglected, and the actual vapor quality for pipes of $\frac{L}{D}$ greater than 12 approached the corresponding long tube value, x_{LT} , in an exponential manner:

$$x_e = x_{LT} \left\{ 1 - \exp \left[-B \left(\frac{L}{D} - 12 \right) \right] \right\} \quad (28)$$

The actual vapor quality at the exit of a long tube was given by

$$x_{LT} = N x_{eq} \quad (29)$$

where N was taken from Equation (26).

By assuming the liquid to be incompressible, and the flow to be frictionless and homogeneous, Henry was able to write the following expression for pressure ratio:

$$r^* = \frac{P_e}{P_o} = 1 - \frac{G_{cr}^2}{P_o} \left[\frac{v_{l,o}}{2C^2} + x_e (v_{g,e} - v_{l,o}) \right] \quad (30)$$

where C is the contraction coefficient at the pipe entrance.

By assuming the vapor compressibility to follow an isothermal process, Henry expressed the critical mass flux as

$$G_{cr}^2 = \frac{1}{\left[\frac{xv_g}{P} - (v_g - v_{l,o}) N \frac{dx_{eq}}{dP} \right]_e} \quad (31)$$

Given a value for B , one can now iterate on Equations (28) through (31) to calculate the exit pressure and the critical mass-flux. From the data of Uchida and Nariai,⁽¹⁷⁾ Henry recommended a value of 0.0523 for the decay constant B . The advantage of this model is that only the test section geometry and the upstream stagnation conditions have to be known. However, no specific recommendation has been given for $\frac{L}{D}$ less than 12.

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For smooth inlet configurations, Henry suggested a value of unity for the contraction coefficient, C , and vapor generation to start from the beginning of the constant area portion. Therefore, Equation (28) was modified as:

$$x_e = x_{LT} \left\{ 1 - \exp \left[- B \left(\frac{L}{D} \right) \right] \right\} \quad (32)$$

where x_{LT} was given by Equation (29) and the same value of B as before was used.

The model showed good agreement with the critical mass-flux data of Fauske,⁽¹⁶⁾ Uchida and Nariai,⁽¹⁷⁾ and Zaloudek.⁽¹⁵⁾ The critical pressure-ratio data were somewhat overpredicted by the model. However, Henry argued that because of the two-dimensional effect near the exit of an abrupt expansion, the pressure taps at the wall recorded lower-than-actual exit (critical) pressures. He also compared the model with Freon-11 and Freon-12 data to justify the use of $(\frac{L}{D})$ as a nonequilibrium scaling parameter.

Henry and Fauske⁽²⁸⁾ extended their nonequilibrium modeling efforts to nozzles, orifices and short tubes. They argued that in a normal nozzle configuration there would be little time for heat and mass transfer in the converging section. Therefore, they assumed that

$$x_t \approx x_o \quad (33)$$

and

$$T_{l,t} \approx T_{l,o} \quad (34)$$

They also assumed the flow to be homogeneous and the expansion up to the throat

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to be isentropic. The vapor compressibility at the throat was, however, assumed to follow a polytropic process such that

$$\left. \frac{dv_g}{dP} \right|_t = - \left. \frac{v_g}{nP} \right|_t \quad (35)$$

where the polytropic exponent, n , was calculated from:

$$n = \frac{(1-x) C_{\ell} / C_{p,g} + 1}{(1-x) C_{\ell} / C_{p,g} + 1/\gamma} \quad (36)$$

The liquid was assumed to be incompressible, and the derivative of slip ratio with respect to pressure at the throat was taken to be zero. The previous definition of nonequilibrium system parameter, N , was invoked and the final expression for critical mass-flux at the throat was written as:

$$G_{cr}^2 = \frac{1}{\left[\frac{x_o v_g}{nP} + (v_g - v_{\ell,o}) \left\{ \frac{(1-x_o) N}{S_{g,eq} - S_{\ell,eq}} \frac{dS_{\ell,eq}}{dP} - \frac{x_o C_{p,g} (1/n - 1/\gamma)}{P (S_{g,o} - S_{\ell,o})} \right\} \right]_t} \quad (37)$$

From the data of Starkman, et al.,⁽¹⁸⁾ Henry and Fauske recommended the following expressions for N for nozzles:

$$\begin{aligned} N &= x_{eq,t}/0.14, & (0 < x_{eq,t} < 0.14) \\ N &= 1, & (x_{eq,t} > 0.14) \end{aligned} \quad (38)$$

From momentum equation, and utilizing Equation (37), they expressed the critical

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pressure ratio as:

$$r^* = \frac{p_t}{p_o} = \left\{ \frac{\frac{1 - \alpha_o}{\alpha_o} (1 - r^*) + \frac{\gamma}{\gamma - 1}}{\frac{1}{2\beta\alpha_t^2} + \frac{\gamma}{\gamma - 1}} \right\}^{\frac{\gamma}{\gamma - 1}} \quad (39)$$

where

$$\beta = \left[\frac{1}{n} + \left(1 - \frac{v_{l,o}}{v_{g,t}} \right) \left(\frac{(1 - x_o) NP_t}{x_o (S_{g,eq} - S_{l,eq})_t} \frac{dS_{l,eq}}{dP} \right)_t - \frac{C_{p,g} (1/n - 1/\gamma)}{(S_{g,o} - S_{l,o})} \right] \quad (40)$$

$$\alpha_o = \frac{x_o v_{g,o}}{(1 - x_o) v_{l,o} + x_o v_{g,o}} \quad (41)$$

$$\alpha_t = \frac{x_o v_{g,t}}{(1 - x_o) v_{l,o} + x_o v_{g,t}} \quad (42)$$

and

$$v_{g,t} = v_{g,o} (r^*)^{-1/\gamma} \quad (43)$$

For given stagnation conditions of P_o and x_o , the transcendental Equation (39) can be solved to determine the throat pressure. The critical mass-flux can then be calculated from Equation (37).

The model showed fair agreement with critical mass-flux data as well as critical pressure ratio data. Comparisons with Starkman et al.'s data⁽¹⁸⁾ are shown in Figures 5a through 5c.

For saturated or subcooled water stagnation condition, i.e., $x_o = 0$, the expression for critical mass-flux [Equation (37)] was written as:

$$G_{cr}^2 = \frac{1}{\left[(v_{g,eq} - v_{l,o}) \frac{N}{S_{g,eq} - S_{l,eq}} \frac{dS_{l,eq}}{dP} \right]_t} \quad (44)$$

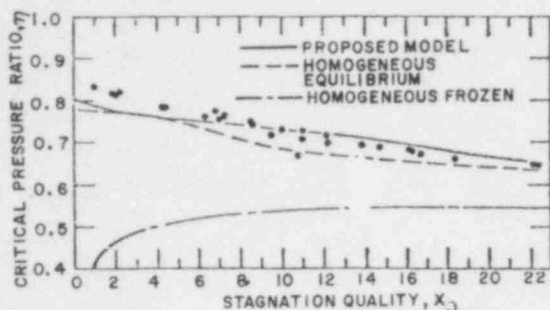


Figure 5a. Comparison of critical pressure ratios(18) with the prediction of Henry and Fauske.(28) (BNL 5-1225-78)

The additional assumption was that the vapor at the throat was saturated at the local pressure. The critical pressure in this case was written as:

$$r^* = \frac{P_t}{P_0} = 1 - \frac{v_{l,0} G_{cr}^2}{2P_0} \quad (45)$$

An iteration on Equations (44) and (45) resulted in the desired critical pressure and critical mass flux. Predictions from this model agreed well with the data of Zaloudek⁽¹⁴⁾ and others.

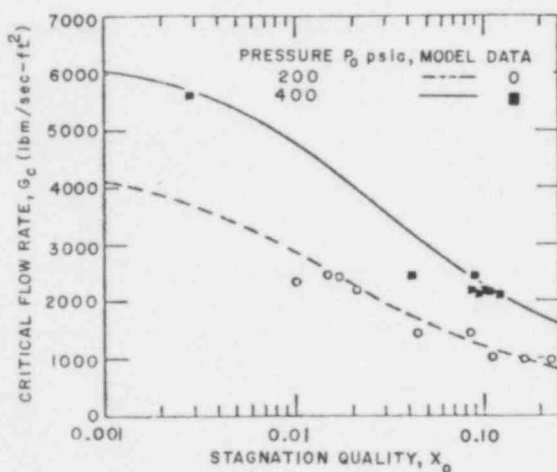


Figure 5b. Comparison of critical flow rates(18) with the prediction of Henry and Fauske.(28) (BNL 5-1226-78)

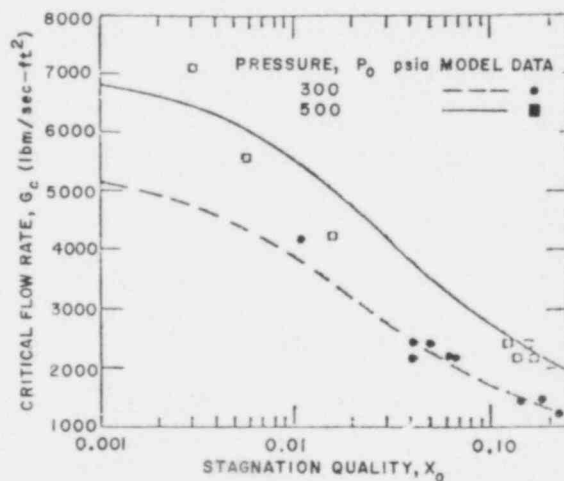


Figure 5c. Comparison of critical flow rates(18) with the prediction of Henry and Fauske.(28) (BNL 5-1228-78)

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For sharp-edged orifices and short tubes with two-phase entrance condition, Henry and Fauske suggested a discharge coefficient of 0.84. This would only modify the expression for critical pressure ratio [Equation (39)] as

$$r^* = \left\{ \frac{\frac{1 - \alpha_0}{\alpha_0} (1 - r^*) + \frac{\gamma}{\gamma - 1}}{\frac{1}{2 (0.84)^2 \beta \alpha_t^2} + \frac{\gamma}{\gamma - 1}} \right\}^{\frac{\gamma}{\gamma - 1}} \quad (46)$$

Equation (37) can then be used to calculate the critical mass-flux.

The above three models of Henry and co-workers^(24,27,28) are useful calculational tools for taking into account the overall effect of thermal non-equilibrium in the prediction of critical mass-fluxes. However, none of these models attempted to describe the mechanism of vapor formation during rapid depressurization or flashing. Moreover, no specific recommendation has been made for pipe lengths less than twelve diameters where a strong length effect can be expected (see Figure 3). Thus, the models could be of limited usefulness in the analyses of a hypothetical LOCA in LWR systems.

5.2 Distributed Models

A large number of distributed models have been developed in recent years. These models have attempted to calculate the actual local rate of vapor generation, and then determine the actual vapor quality at the critical location. We shall discuss these models in detail.

Edwards⁽²⁹⁾ attempted to formulate the problem from a mechanistic viewpoint. The major assumptions of his model were:

- (a) homogeneous flow
- (b) the vapor phase was always at saturation
- (c) all bubble nuclei were formed simultaneously following an arbitrary

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- time delay corresponding to a certain degree of liquid superheating
- (d) for sharp-edged entries, causing initial flow separation, downstream flow expansion took place at constant pressure
 - (e) results of Plesset and Zwick⁽³⁰⁾ for bubble growth at constant pressures could be applied during the initial phase of bubble growth
 - (f) at the later stage, when the pressure was falling, the vapor generation could be calculated from a one-dimensional composite slab heat conduction model
 - (g) the surface area of vapor-liquid interface per unit mass of liquid could be approximated by:

$$A_b = (4\pi N)^{1/3} \left[3x (1 - x) \frac{v_l v_g}{v} \right]^{2/3} \quad (47)$$

where N is the number of bubble nuclei per unit mass of liquid, x is the actual vapor quality and v is the specific volume of the mixture.

Based on the above assumptions, a set of one-dimensional conservation equations was solved to calculate the pressure, velocity and the specific volume along the length of a pipe. The flow was taken to be choked when the pressure gradient at the exit of the pipe became infinite. However, this choking criterion does not agree with the finite experimental pressure gradient obtained by Reocreux.^(43,44)

There were two arbitrary constants in the above model. These were:

- (1) the time-delay for bubble nucleation which was on the order of 1 millisecond,
- (2) the number of bubble nuclei per pound of liquid which ranged from 10^8 to 10^{11} .

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Utilizing the data of Fauske⁽¹⁶⁾ taken in 6.35 mm diameter tubes, Edwards suggested the following correlations for those two items:

$$(i) \text{ Time delay: } \Delta t_E = C \exp \left\{ -B \frac{(P_{sat} - P)}{P_{sat}} \right\} \quad (48)$$

$$\text{where } C = F(\sigma) = 2.85 \times 10^{-4} \exp(7.2 \times 10^7 \sigma^3) \quad (49)$$

$$\text{and } B = 4.605 \quad (50)$$

$$(ii) \text{ Bubble nuclei: } N = 3.0 \times 10^{11} \exp \left\{ -190 \left(\frac{\sigma}{P_{sat} - P} \right)^{1/3} \right\} \quad (51)$$

(per pound of water).

Pressures and surface tensions are expressed in psi and lbf/ft, respectively, in the above correlations. For short tube lengths ($\frac{L}{D} < 5$), an arbitrary radial pressure gradient near the exit was needed to match the data. This was necessary because of the abrupt expansion at the exit.

Applicability of the above model to a large diameter pipe was checked by Edwards and O'Brien.⁽³¹⁾ Several transient depressurization tests were carried out in a steel pipe, 2.88" (73 mm) in inside diameter and 13.44 ft. (4097 mm) in length. The pipe was filled with subcooled water at high pressures (34.5 to 172.4 bar). Transients were initiated by rupturing a disc at one end of the pipe. During the transient, pressures at the wall were recorded along the length of the pipe. These pressure traces including a dip during the early part of the transient could be well predicted by Edwards' model by choosing a suitable value for the number of bubble nuclei.* However, the prediction did not agree well with the long term pressure traces. An error in the computer program was

*The time delay had only a small effect.

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blamed for the discrepancy. In spite of this, it is clear that although the model showed some promise yet the correlations for the time delay and the number of bubble nuclei [Equations (48) and (51)] cannot be considered as general.

Malnes⁽³²⁾ emphasized the importance of dissolved gas in water by taking its effect into account in his critical flow model. By assuming a homogeneous flow model and $1 - \alpha(1 - \rho_v/\rho_\ell) \approx 1$, he expressed the mass rate of vapor generation per unit volume as:

$$\Gamma_v = \rho_v \frac{D\alpha}{Dt} \quad (52)$$

where $\frac{D}{Dt} = \frac{\partial}{\partial t} + u \frac{\partial}{\partial z}$.

Like Edwards, Malnes used the conduction-controlled bubble growth law applicable to a liquid of constant pressure and constant superheat. He also assumed that the number of bubbles per unit volume was constant, and would be a function of $(g\rho_\ell/\sigma)$. Finally, by combining the vaporization due to bubble growth and flashing from droplets, Malnes expressed the vapor generation rate as:

$$\Gamma_v = (R_0 \frac{\rho_v}{\rho_\ell} F + R_1) \frac{C_p k_\ell g \rho_\ell}{\sigma h_{fg}^2} \{ \alpha(1 - \alpha) \}^{1/3} (T_\ell - T_{sat})^2 \quad (53)$$

where R_0 and R_1 are two dimensionless empirical constants, and

$$\begin{aligned} F &= 1 - 2\alpha & (\alpha < 0.5) \\ F &= 0 & (\alpha > 0.5) \end{aligned} \quad (54)$$

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A set of conservation equations were solved along with Equation (53) along the length of a pipe, and inlet mass-flux was increased until critical velocity was obtained at the exit of the constant diameter pipe. Values of 7×10^7 and 2×10^5 for the constants R_0 and R_1 , respectively, were obtained from the best fit of Henry's data,⁽²⁵⁾ where exit void fractions were measured. Gas content of the order of 0.01 to $0.06 \text{ m}^3/\text{m}^3$ at 1 bar and 25°C was assumed. This large gas content was necessary to explain the observed void fractions before the pressure reached saturation. Finally, good agreement with Henry's critical mass flux data was shown by assuming an average gas content of $0.04 \text{ m}^3/\text{m}^3$ at 1 bar and 25°C . Malnes pointed out that nitrogen gas was used to pressurize the water tank in Henry's experiment, and therefore, a large amount of dissolved gas in water could be expected.

To match Zaloudek's data,⁽¹⁵⁾ Malnes needed a relatively small ($0.5 \times 10^{-3} \text{ m}^3/\text{m}^3$ at 1 bar and 25°C , typical for BWR) amount of gas. The predictions with the same amount of dissolved gas also agreed well with Fauske's critical flow rate data⁽¹⁶⁾ for tube lengths greater than 5 cm (i.e., $\frac{L}{D} > 8$). However, for shorter tube lengths the predictions were much lower than the data. Malnes attributed this discrepancy to the phenomenon of delayed flashing which was not included in his model. Similar discrepancy was found even with respect to the long tube length data of Uchida and Nariai,⁽¹⁷⁾ and the same explanation was offered.

Rohatgi and Reshotko⁽³³⁾ made an attempt to relax the assumption of a constant number of bubble nuclei per unit mass of liquid as employed by Edwards.⁽²⁹⁾ Following the kinetic theory of bubble nucleation in a superheated liquid, they wrote the following expression for the rate of production

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of bubble nuclei per unit volume of liquid:

$$\frac{dN_l}{dt} = (HN) \sqrt{\frac{2\sigma}{\pi m}} \exp \left[- \frac{16\pi\sigma^3}{3 (P_v - P_l)^2} \frac{\phi}{KT_l} \right] \quad (55)$$

where HN is the number of heterogeneous nucleation sites per unit volume of liquid, m is the mass of a molecule, K is the Boltzmann constant and ϕ is a contact angle function given by

$$\phi = \frac{(2 - \cos\theta)(1 + \cos\theta)^2}{4} \quad (56)$$

Plesset and Zwick's formulation⁽³⁰⁾ for the bubble growth was assumed, and the bubbles were tracked downstream from their location of nucleation. Finally, the vapor void fraction at any location, Z, was given by

$$\alpha = \frac{4\pi}{3} \int_{Z_0}^Z \left[r_0 + \int_{Z_1}^Z \frac{dr}{dZ_2} dZ_2 \right]^3 \frac{dN_l}{dZ_1} [1 - \alpha(Z_1)] dZ_1 \quad (57)$$

where Z_0 represents the saturation location, and r_0 is the initial bubble radius at nucleation site Z_1 .

Rohatgi and Reshotko used Simoneau's data⁽³⁴⁾ taken in a converging-diverging nozzle with liquid nitrogen as the operating fluid to recommend values for ϕ and HN. A very small value (5×10^{-6}) for ϕ , which corresponds to a value of approximately 176° for contact angle θ , was required to match the data. The value of HN ranged from 1 to 2 per cm^3 . This was a very small value compared to the bubble nuclei number proposed by Edwards.⁽²⁹⁾ However, no comparison was made with any steam-water data, and the model did not include the effect of delayed flashing.

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As seen from Equation (53), a conduction-controlled bubble growth law at constant pressure, although not realistic during a flashing flow, results in a vapor generation rate which is proportional to the square of the liquid superheating. In a different type of formulation, Rivard and Torrey⁽³⁵⁾ assumed the vapor generation rate to be proportional to the liquid superheating, and wrote the following expression:

$$\Gamma_v = \lambda_e A(1 - \alpha)\alpha \rho_\ell (T_{\text{sat}} R_u)^{1/2} (T_\ell - T_{\text{sat}})/T_{\text{sat}} \quad (58)$$

where λ_e is an arbitrary constant, R_u is the universal gas constant and A is proportional to the vapor-liquid interfacial area per unit mixture volume. For N equal sized spherical bubbles (or droplets) per unit volume, A is given by:

$$A = \alpha^{2/3} \left(\frac{4\pi N}{3}\right)^{1/3} \quad (\alpha \leq 0.5) \quad (59)$$

$$A = (1 - \alpha)^{2/3} \left(\frac{4\pi N}{3}\right)^{1/3} \quad (\alpha \geq 0.5)$$

Two-phase flow equations corresponding to the two-fluid model, i.e., six conservation equations, were solved by the computer code called KACHINA.⁽³⁰⁾

Edwards and O'Brien's transient data⁽³¹⁾ were used for model comparison. Values of 0.1 for λ_e and 10 bubbles per cm^3 for N seemed to predict the 1000 psig (70 bar) data with limited success. Rivard and Torrey also reported that the values for λ_e and N had some effect during early period of the transient, but no significant effect at a later stage. This indicates the importance of non-equilibrium effects when initially subcooled liquid is depressurized rapidly.

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Wolfert⁽³⁷⁾ calculated the rate of vapor generation by allowing a relative velocity between the vapor bubbles and the liquid. He started from the following expression for heat transfer coefficient, h_ℓ , between the bubbles and the liquid:

$$Nu_b = \frac{h_\ell 2r_b}{k_\ell} = \frac{2}{\sqrt{\pi}} \left(\frac{2r_b \rho_\ell c_{p,\ell} u_r}{k_\ell} \right)^{1/2} = \frac{2}{\sqrt{\pi}} (Pe_b)^{1/2} \quad (67)$$

where u_r is the relative velocity.

Wolfert finally expressed the rate of vapor generation per unit volume for the case of bubbles with relative velocity as:

$$\Gamma_{v,u_r \neq 0} = (24Nu_r \alpha)^{1/2} (k_\ell \rho_\ell c_{p,\ell})^{1/2} (T_\ell - T_{sat})/h_{fg} \quad (61)$$

where N is the number of bubbles per unit mixture volume. Notice that the vapor generation rate in this case is proportional to the liquid superheating, and in this respect, somewhat similar to the formulation of Rivard and Torrey, i.e., Equation (58).

Wolfert also considered the case with no relative velocity, and starting from the bubble growth law of Plesset and Zwick,⁽³⁰⁾ he finally obtained:

$$\Gamma_{v,u_r=0} = 24 \left(\frac{3}{4\pi} \right)^{1/3} N^{2/3} \alpha^{1/3} \frac{(k_\ell \rho_\ell c_{p,\ell})}{\rho_v} \frac{(T_\ell - T_{sat})^2}{h_{fg}^2} \quad (62)$$

Therefore, the functional relationship between the vapor generation rate and the liquid superheating clearly depends on whether the relative velocity is considered or not. A comparison between Equations (61) and (62) showed that at small vapor void fraction, particularly at high liquid superheating,

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$\Gamma_{v,u_r=0}$ dominates, whereas at higher void fraction the vapor generation rate would be determined primarily by $\Gamma_{v,u_r \neq 0}$. Wolfert just added the two terms without justification and expressed the vapor generation rate per unit volume as

$$\Gamma_v = \Gamma_{v,u_r=0} + \Gamma_{v,u_r \neq 0} \quad (63)$$

Wolfert applied the model to predict Edwards and O'Brien's 1000 psig (70 bar) test data.⁽³¹⁾ A value of 0.15 m/sec was used for the relative velocity of the bubbles, and a value of 5×10^9 per m^3 for number of bubbles gave reasonable agreement with the test data. It should be noticed that Rivard and Torrey⁽³⁵⁾ used a bubble density of only 10^7 per m^3 to match the same data, whereas Edwards⁽²⁹⁾ recommended bubble densities in the range of 2×10^{11} to 2×10^{14} per m^3 .

Wolfert also compared the above model with the data of Friz, et al.⁽³⁸⁾ In that experiment, a fixed volume of subcooled water at high pressure was suddenly expanded to a slightly larger volume and the pressure-time history was recorded. After a number of trials and errors, Wolfert recommended the combined form of the vapor generation rate, i.e., Equation (63), with a bubble density of 10^9 per m^3 . The comparison of the data with various values of bubble densities is shown in Figure 6. It should be noticed that the pressure dip below the saturation pressure was a marked manifestation of the thermal non-equilibrium effects during a sudden expansion.

Friz, et al.,⁽³⁸⁾ themselves made an attempt to predict their experimental data. They used the Plesset and Zwick bubble growth law at constant pressure⁽³⁰⁾ and derived an expression for vapor generation rate, very similar to Equation (62).

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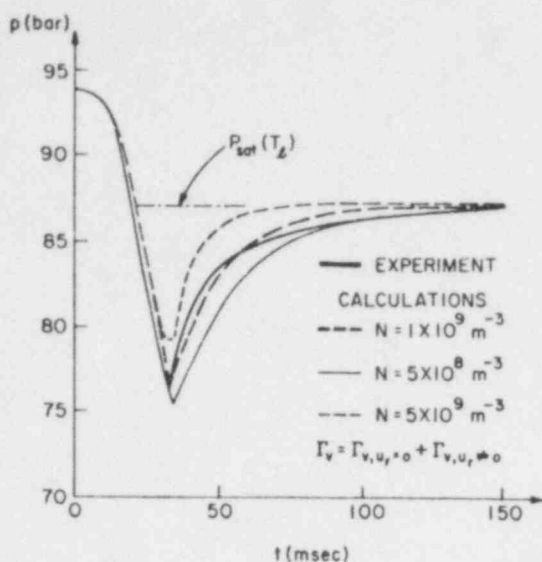


Figure 6. Comparison of the prediction of Wolfert⁽³⁷⁾ with the data of Friz, et al.⁽³⁸⁾ (BNL 5-1229-78)

They found the bubble density to be a strong function of water temperature, and recommended bubble density values in the range of 10^8 to 2×10^{10} per m^3 for water temperatures of $250^\circ C$ to $325^\circ C$. However, they recognized that the bubble density, which might be influenced by impurities, gas content and wall effects, was the most uncertain factor in their theoretical model.

In contrast to the rather "ad

hoc" forms of the vapor generation rate described until now, Bouré, et al.,⁽³⁹⁾ suggested that a general constitutive equation for mass transfer rate between the phases should include derivatives of the dependent variables. The reason for this suggestion was that under some circumstances imaginary characteristics were encountered and only the terms containing the derivatives would enter the system determinant (which must vanish at the critical section). Therefore, only the differential terms of the constitutive equation could possibly have the effect of rendering real characteristics under all conditions, and have an effect on the critical flow. Liles⁽⁴⁰⁾ has attempted to follow the above suggestion to formulate the constitutive equation for vapor generation, but no correlation has been proposed.

In an attempt to compare various forms of the constitutive equation, Kroeger⁽⁴¹⁾ studies three different expressions. The first expression was the same as the vapor generation rate corresponding to the homogeneous equilibrium

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model:

$$\Gamma_I = \Gamma_{HEM} = \frac{\rho_m}{h_{fg}} \left\{ \frac{1}{\rho_m} - \left[(1 - c) \frac{dh_f}{dP} + c \frac{dh_g}{dP} \right] \right\} \frac{DP}{Dt} \quad (64)$$

where $c (= \alpha \rho_v / \rho_m)$ is the mass concentration of vapor, and the substantial derivative is taken with respect to the mixture velocity, $u_m (= G / \rho_m)$. The second expression was a delayed relaxation model for nonequilibrium, nonhomogeneous flashing flow:

$$\Gamma_{II} = \begin{cases} 0 & (c_{eq} \leq c_A) \\ \Gamma_{HEM} \left[1 + \frac{c_{eq} - c}{\tau^*} \right] & (c_{eq} > c_A) \end{cases} \quad (65)$$

where c_A was a threshold value of the equilibrium vapor mass concentration, c_{eq} , up to which no vapor generation was assumed. Vapor generation was assumed to start at $c_{eq} = c_A$ at a rate larger than Γ_{HEM} , and finally relaxed to the equilibrium state, controlled by the non-dimensional time constant τ^* . This expression also implies that for $c_{eq} > c_A$, Γ_{II} must always be greater than Γ_{HEM} which may not be realized in practice.

The third expression was an alternate relaxation type model similar to that of a first-order chemical reaction⁽⁴⁸⁾

$$\Gamma_{III} = \rho_m \frac{c_{eq} - c}{\tau} = \rho_m \frac{(1 - c) (i_g - i_f)}{\tau h_{fg}} \quad (66)$$

where τ was the time constant. Notice that the vapor generation rate according to the above equation is approximately proportional to the liquid superheating, and therefore, similar to those used by Rivard and Torrey [Equation (58)], and

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Wolfert [Equation (61)]. Also notice that the first two expressions used by Kroeger, i.e., Equations (64) and (65), contain derivatives of dependent variables, as suggested by Bouré, et al., whereas the third expression, i.e., Equation (66), contains no such derivative. Equation (65) also contains a nucleation delay as suggested by Edwards.⁽²⁹⁾

All of the above three expressions for the vapor generation rate were used in a drift-flux formulation of the transient tests of Edwards and O'Brien.⁽³¹⁾ Method of characteristics was used as the solution procedure and the condition of no backward characteristic (similar to the vanishing condition of the system determinant) was employed as the choking criterion. Unfortunately, the resultant pressure-time histories were found to be quite insensitive of the chosen constitutive relations, and all the predictions agreed reasonably well with the test data. Kroeger, therefore, concluded that the tests of Edwards and O'Brien⁽³¹⁾ were not suitable experiments for determining the effects of thermal nonequilibrium during a flashing flow.

Bauer, et al.,⁽⁴²⁾ incorporated the effect of thermal nonequilibrium by implying the following equation for the actual vapor quality:

$$\frac{\partial x}{\partial t} + u \frac{\partial x}{\partial z} = \frac{\Gamma_v}{\rho_m} = \frac{x_{eq} - x}{\tau} \quad (67)$$

where u is the homogeneous flow velocity and τ is a time constant. The following correlation for this time constant was evaluated from MOBY DICK tests^(43,44):

$$\tau = 660 P^{-0.505} u^{-1.89} \alpha^{-0.954} \quad (68)$$

where τ is in seconds, P is in N/m^2 and u is in m/sec . The range of conditions

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covered was:

$$\begin{aligned} 1.2 &< P < 8 \text{ bars} \\ 5 &< u < 54 \text{ m/sec} \\ 0.01 &< \alpha < 0.96 \\ 0.001 &< \tau < 1 \text{ sec} \end{aligned}$$

However, no theoretical basis for the correlation for τ , i.e., Equation (68), was given. Therefore, one should be cautious in extrapolating the correlation beyond the range of its data base.

Jones⁽⁴⁹⁾ has shown that Bauer, et al's formulation was indeed correct for steady, one-dimensional flow. In addition, he showed that the constitutive law for the rate of vapor generation may be given as

$$\Gamma_v = \rho_m \frac{x_{eq} - x}{\tau} = \Gamma_{eq} Nr_f (x_{eq} - x) \quad (69)$$

which shows that the relaxation time constant, τ , may be expressed in terms of a flashing relaxation number, Nr_f , as

$$\tau = \frac{\rho_m}{\Gamma_{eq} Nr_f} \quad (70)$$

where Γ_{eq} is the equilibrium rate of vapor generation. The flashing relaxation number, Nr_f , may be expressed in terms of the net interfacial heat flux and interfacial area density, and must be correlated against actual measurements. This has not yet been accomplished. It may be noted that the nonequilibrium

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evaporation rates may be either smaller or larger than the equilibrium values, the former in the early stages and the latter as equilibrium is being approached along the pipe.

In summary, there are basically four types of models to take into account the effects of thermal nonequilibrium. In the first type of models, namely those of Henry and co-workers,^(24,27,28) direct assumptions regarding the actual vapor quality and various derivatives at the critical location are employed. Although these models are useful calculational tools, their applicability to transient blowdown situations has yet to be proved. The second type of models, i.e., those of Edwards,⁽²⁹⁾ Malnes,⁽³²⁾ and others, takes a more mechanistic approach towards the problem. Some of the models have attempted to consider the nucleation delay, the nucleation rate, the bubble growth law, the bubble population, and even the effect of dissolved gas. However, at present, there is no general correlation for any of these items, and further work is necessary in this area. In particular, the present models use the bubble growth law at constant pressure, which could be in error if the bubble transit time through the pipe is not extremely small.⁽⁵⁰⁾ The third class of models, suggested by Bouré, et al.,⁽³⁹⁾ incorporates terms containing derivatives of dependent variables in the expression for vapor generation. However, no widely accepted functional form of such an expression is available yet. Finally, the relaxation-type models as suggested by Bauer, et al.,⁽⁴²⁾ and Jones⁽⁴⁹⁾ seem to show promise in describing the nonequilibrium vapor generation rate in a simple, and yet phenomenological way. Further work is still needed in representing these models with the interfacial heat transfer rate and the interfacial area characteristics of the flow field.

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As for the choking criterion, the vanishing condition of the system determinant is a necessary one. However, it is not a sufficient condition. Several compatible conditions must also be satisfied for the flow to be "critical." These have been discussed in a recent paper by Bouré.⁽⁵¹⁾ Interested readers are urged to read the paper.

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6. DISCUSSION

With the exception of James',⁽⁴⁵⁾ study, all the other steady-state critical flow experiments have been conducted in small diameter (< 30 mm) pipes and nozzles. Although some transient tests have been carried out in intermediate pipe diameters (~ 100 mm), there is no data (transient or steady-state) in reactor-size pipe diameters (~ 300 mm) with relatively short flow lengths.* In small diameter pipes, however, experimental data are available in various pipe lengths (short as well as long) with a wide range of vapor qualities and pressures.

A comparison of various experimental data (as shown in Table 1) reveals that there are disagreements regarding the effect of pipe diameters on the critical flow rates. Although most of the experiments do not show any significant effect of pipe diameters, there is evidence^(21,47) which suggests that the critical mass-flux decreases with increasing pipe diameters. The reason for this anomaly is not understood at this time. However, there is no discrepancy regarding the effect of pipe length. All experiments show that the critical flow rate increases significantly as the pipe length is shortened, and brought near zero. Critical flow rate also increases with decreasing upstream enthalpy, particularly for near-saturation and subcooled liquid conditions. Both of these effects, have been attributed to the thermal nonequilibrium phenomena associated with a bubble nucleation delay and a limited, heat transfer controlled, rate of vapor generation. The amount of dissolved gas in liquid, nucleation characteristics of the wall and the entrance geometry for a short pipe are also believed to have an effect on the degree of thermal nonequilibrium, and thus affect the critical flow rate.

* Large-scale (pipe diameter ~ 500 mm) transient critical flow experiments have recently begun in Marviken test apparatus in Sweden.

Although the pipe length has a strong effect on the critical flow rate for short pipes, it is not yet established whether the pipe length, L , or the pipe length-to-diameter ratio, L/D , or both should be the governing parameter(s). It has been suggested^(21,22) that the nonequilibrium effect is primarily due to the length of the pipe, and becomes insignificant if the pipe length exceeds 127 mm. On the other hand, some researchers⁽²⁷⁾ have used the pipe length-to-diameter ratio, L/D , as the governing parameter for thermal nonequilibrium. However, there are experimental findings^(23,47) which disagree with both of the above viewpoints. Therefore, it is possible that both the length and the diameter, along with the upstream fluid conditions, are important in determining the critical flow rates.

The simplest analytical model, namely the isentropic Homogeneous-Equilibrium Model (HEM), underpredicts the critical flow rate data considerably for short pipe lengths and low upstream qualities, including the subcooled liquid condition. The reason for this discrepancy is mainly due to the model's inability to account for the thermal nonequilibrium effects. The model also underpredicts the critical flow rate data, particularly at low system pressures, mainly because of the homogeneous flow assumption of the model. Therefore, this model can serve as the lower bound of the actual critical flow rates, and is acceptable only for high pressures, relatively long pipes and relatively high upstream vapor qualities.

The effect of relative velocity has been considered in a number of models, namely those of Fauske,^(6,7) Moody⁽¹⁰⁾ and Levy.⁽¹²⁾ Theoretical treatments of these models are not completely correct, and the experimental data on relative velocities show much lower values than those taken in these models. Like the homogeneous-equilibrium model, these models also underpredict the critical flow rate data for short pipes and low upstream qualities. Therefore, the Moody

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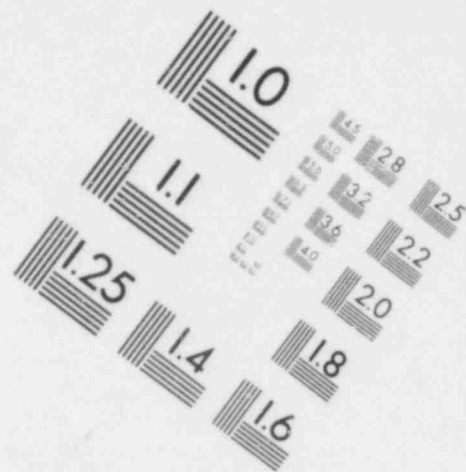
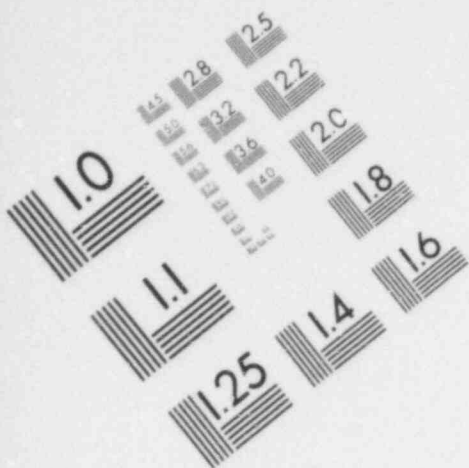
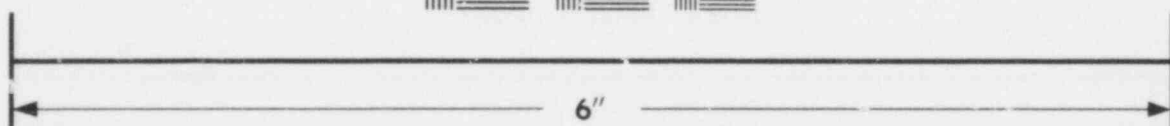
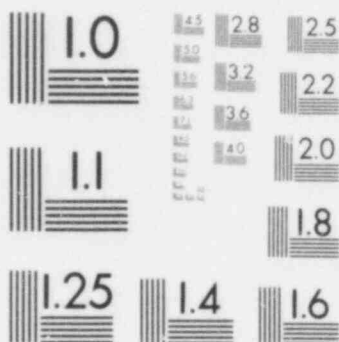
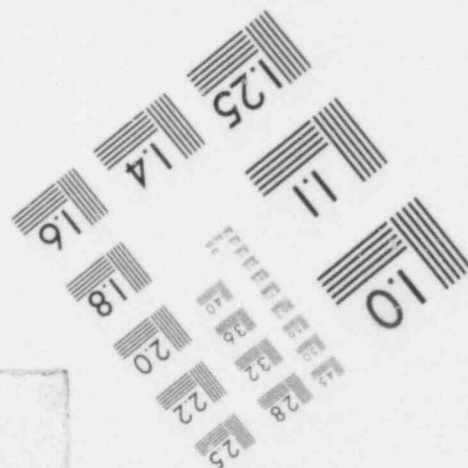
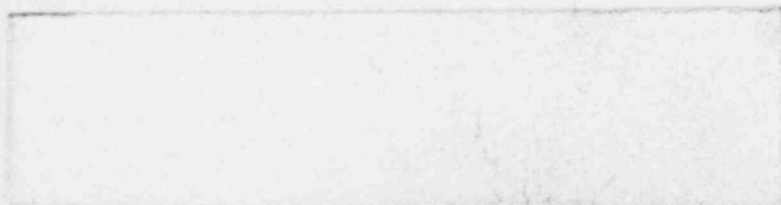
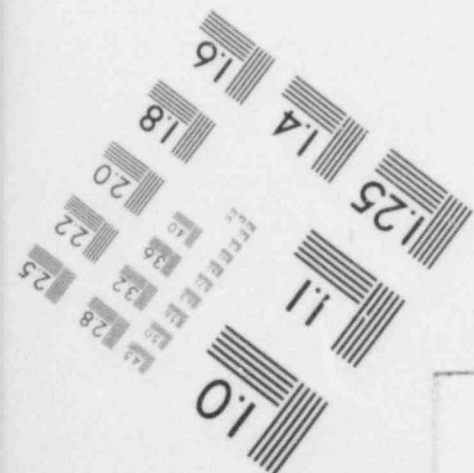


IMAGE EVALUATION TEST TARGET (MT-3)



MICROCOPY RESOLUTION TEST CHART



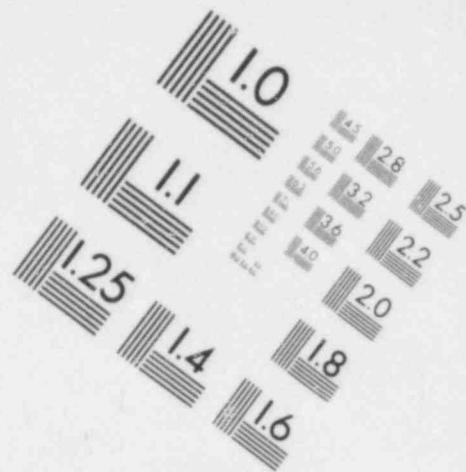
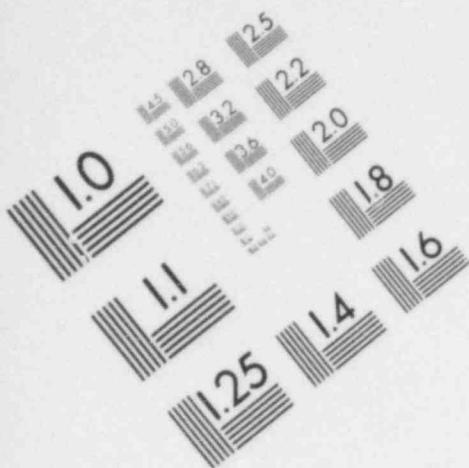
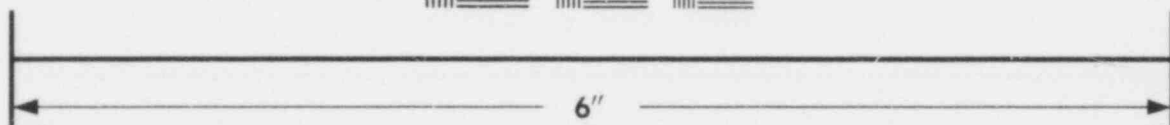
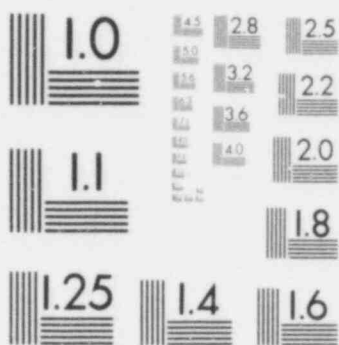
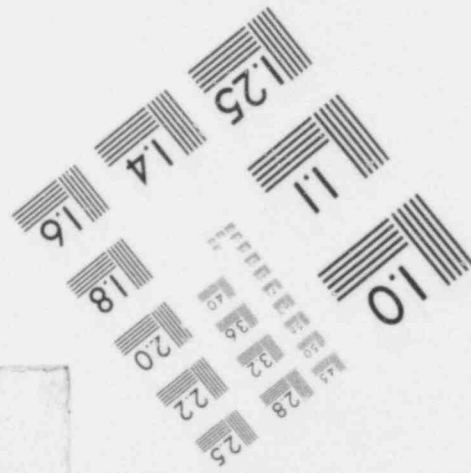
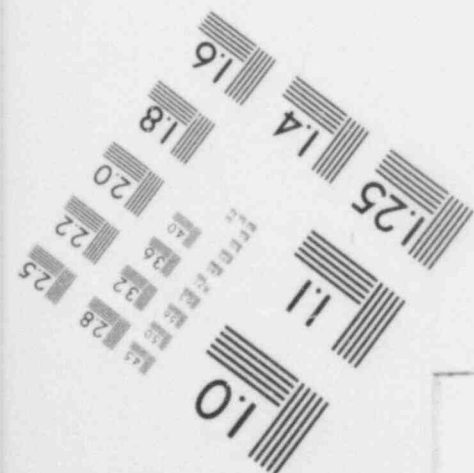


IMAGE EVALUATION TEST TARGET (MT-3)



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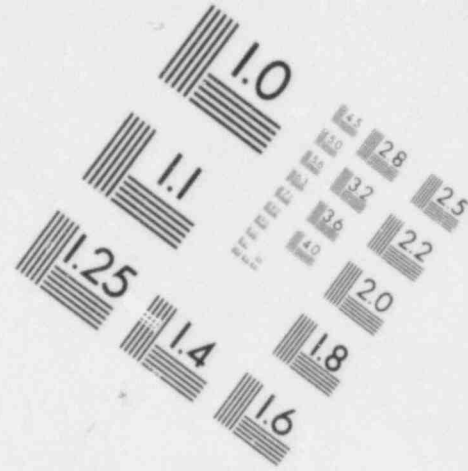
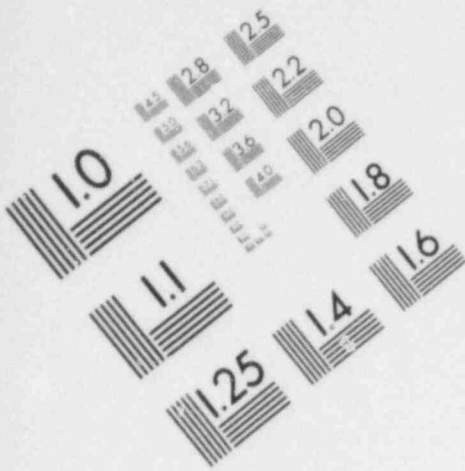
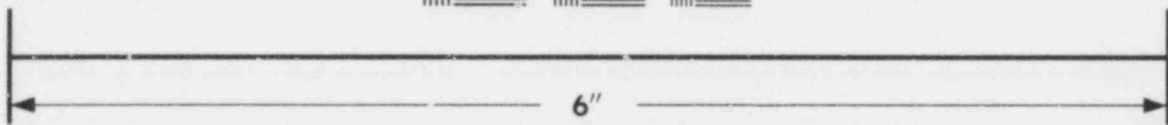
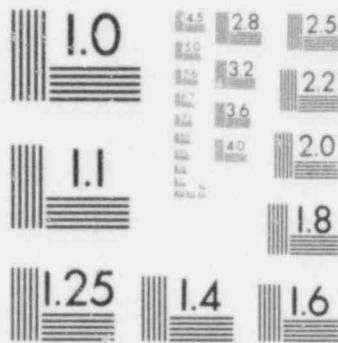
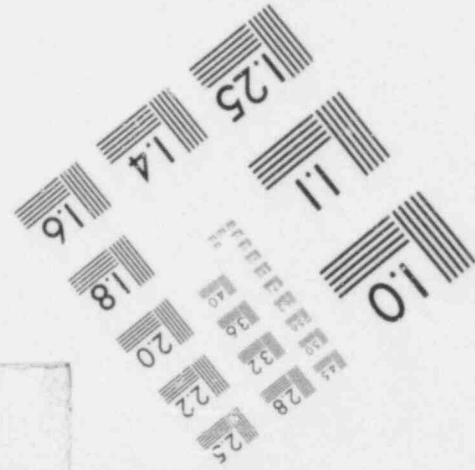
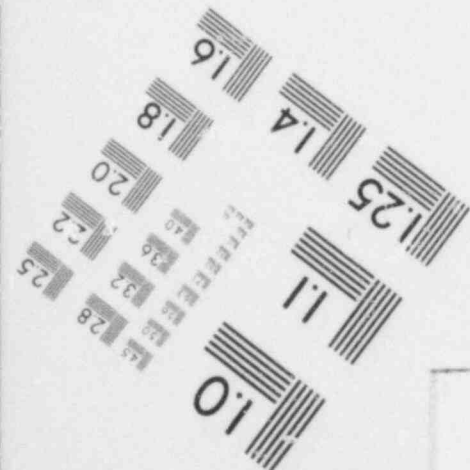


IMAGE EVALUATION TEST TARGET (MT-3)



MICROCOPY RESOLUTION TEST CHART



model⁽¹⁰⁾ which is approved for LWR licensing calculations, may underpredict the break flow rate during the early stages of a hypothetical LOCA. Uncertainties in the break flow rate can then alter the transient core thermal-hydraulic calculations.

It is generally agreed upon that the effect of thermal nonequilibrium must be taken into account for critical flow rates in a short pipe with low upstream qualities including the subcooled liquid condition. Such is the case during the early stages of a hypothetical LOCA in a LWR system. Therefore, a number of models have been proposed in recent years to include the effect of thermal nonequilibrium. These can be categorized as (i) lumped models, and (ii) distributed models. The lumped models, as developed by Henry and co-workers,^(24,27,28) do not include the length effect for pipe lengths less than twelve diameters where a strong length effect has been found experimentally.⁽¹⁶⁾ Moreover, because of the nonmechanistic nature of these lumped models, it is questionable whether they can be applied to transient situations with confidence. On the other hand, the distributed models^(29,32,33,37,42) have attempted to formulate the problem from a mechanistic viewpoint. Some great efforts have been made in quantifying the bubble nucleation delay,⁽²⁹⁾ the rate of bubble nucleation⁽³³⁾ and the bubble population density. However, no general correlation for any of these items is available yet. In view of the complexity of the problem, it seems that the relaxation-type models, as proposed by Eauer, et al.,⁽⁴²⁾ and Jones,⁽⁴⁹⁾ are the logical approaches to take at this time. However further work is needed to develop phenomenological correlations for this type of models.

Because of the influence of the entrance geometry and the sharp pressure gradient at the critical section (exit section for a constant diameter pipe) it

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is expected that a distributed formulation will be superior to a lumped approach. At the critical section, the system determinant must vanish. However, for the flow to be "critical" several compatible conditions must also be satisfied. These have been discussed by Boure⁽⁵¹⁾ in detail.

Finally, a two-phase critical flow model which is applicable to the entire range of expected conditions during a hypothetical LOCA in a LWR system, is not available yet. Such a model must include the effects of thermal nonequilibrium as well as relative velocity between the phases. The effect of dissolved gases and other parameters which significantly affect the vapor-liquid interfacial area densities and the rate of heat transfer to the interfaces, must be included. The model should be able to predict the all liquid Bernoulli flow at one end (subcooled liquid discharging through an orifice), and approach the homogeneous-equilibrium model at the other end (high quality two-phase mixture discharging through a long pipe). Such an effort, combined with controlled experiments to measure the actual rate of vapor generation, is currently under way at Brookhaven National Laboratory.

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7. CONCLUSIONS AND RECOMMENDATIONS

Based on the previous discussion, the following conclusions can be drawn:

1. The Homogeneous-Equilibrium Model (HEM) underpredicts the critical flow rates for short pipes and near saturation or subcooled upstream conditions.
2. The equilibrium slip models of Fauske,^(6,7) Moody⁽¹⁰⁾ and Levy,⁽¹²⁾ although successful for long tubes, underpredict the critical flow rates for short pipes. This is particularly true if the upstream condition is subcooled or near saturation.
3. Effects of thermal nonequilibrium must be taken into account for short pipes. However, it is not clear whether the pipe length, L , or the pipe length-to-diameter ratio, L/D , or both are important in determining the effects of thermal nonequilibrium.
4. At present, there is no general model or correlation for critical flow which is valid for a wide range of pipe lengths, pipe diameters, and upstream conditions including subcooled liquid.

Based on the above conclusions, the following recommendations seem to be appropriate:

1. A general model for critical flow, which considers both the thermal nonequilibrium and the relative velocity between the phases, must be developed.
2. In view of the complexity of two-phase flashing flow, a relaxation-type nonequilibrium model seems to be the logical approach at this time.
3. The general model must include the effects of dissolved gases, the heterogeneous nucleation characteristics of the wall, and other

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parameters which affect the interfacial area densities, and thus affect the rate of vapor generation (or the degree of thermal non-equilibrium).

4. The general model must be able to predict the all liquid Bernoulli flow at one end and the homogeneous-equilibrium flow at the other end.
5. Controlled experimental data must be obtained in various pipe diameters (including reactor sizes, if possible), pipe lengths and operating conditions so that the general model can be verified for all possible conditions.

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